

Data for Policy Analysis and Management

CLASS 2: DATA,
MEASUREMENT AND
VARIABLES

Instructor: Aida Pacheco-Applegate
Summer 2026

Agenda

1. Research Questions
 - Small Group Exercise
2. Descriptive vs. Inferential Statistics
3. Variables
 - Small Group Exercise
4. Lab Session: Describing Variables

What Is a Research Question?



A research question is a question that:

- Can be answered with empirical evidence.
- Improves understanding of how and why the world works.



Grounded in key concepts:

- Theory = explanation of why something happens.
- Hypothesis = testable prediction derived from theory.



A strong research question:

- Is linked to a theory.
- Helps test, refine, or challenge that theory.

Key Features

Where do research questions come from?

- Curiosity → How does something work? Why?
- Theories/Hypotheses → What should we observe if this idea is true?
- Opportunity → What can available data help us understand?

What makes a “good” research question?

- Simple and specific
- ✗ “What are the determinants of social mobility?”
- ✓ “Is birth location a determinant of social mobility?”
- Answerable
 - Does the necessary data exist? Is the information knowable?
- Feasible
 - Can you realistically collect the data (time, cost, access)?
- Empirically relevant
 - Would the answer be useful or meaningful?



Types of Questions and the Limits of Data

Types

Descriptive (what is happening?)

- % of children ages 0–2 enrolled in ECE programs in Illinois

Predictive (relationships between variables)

- Are children ages 3–5 more likely to be enrolled in ECE programs than children ages 0–2?

Causal (what causes what?)

- Does the federal child care subsidy program cause children ages 0–2 in Illinois to enroll in ECE programs?

Data Mining

Data mining refers to conducting research by directly analyzing existing data without relying on a research question.

Good for:

- Finding patterns
- Making predictions

But limited:

- Correlation $\neq$ causation
- Does not explain why
- Risk of false positives

Example:

Ice cream sales $\uparrow$ and shorts-wearing $\uparrow$
but both driven by temperature!

Key Takeaway

Theory $\rightarrow$ *Question* $\rightarrow$ *Hypothesis* $\rightarrow$ *Data* $\rightarrow$ *Explanation*

Small Group Exercise

Directions

Form a group of 4 and develop one research question that can be answered using quantitative data and is related to social policy. You will have **10 minutes** to complete the exercise.

Then, briefly answer the following:

- Research question: What is the question you want to answer?
- Type of research question: Is it descriptive, causal, or predictive? Briefly justify your choice.
- Data needed: What type of quantitative data would you use (e.g., administrative records, survey data, panel data, census data)? Who or what would be included in your dataset?



Descriptive vs. Inferential Statistics

Population vs. Sample

- Population = entire group of interest (e.g., all U.S. adults).
- Sample = subset observed in a study (e.g., 1,326 survey respondents).

Descriptive Statistics

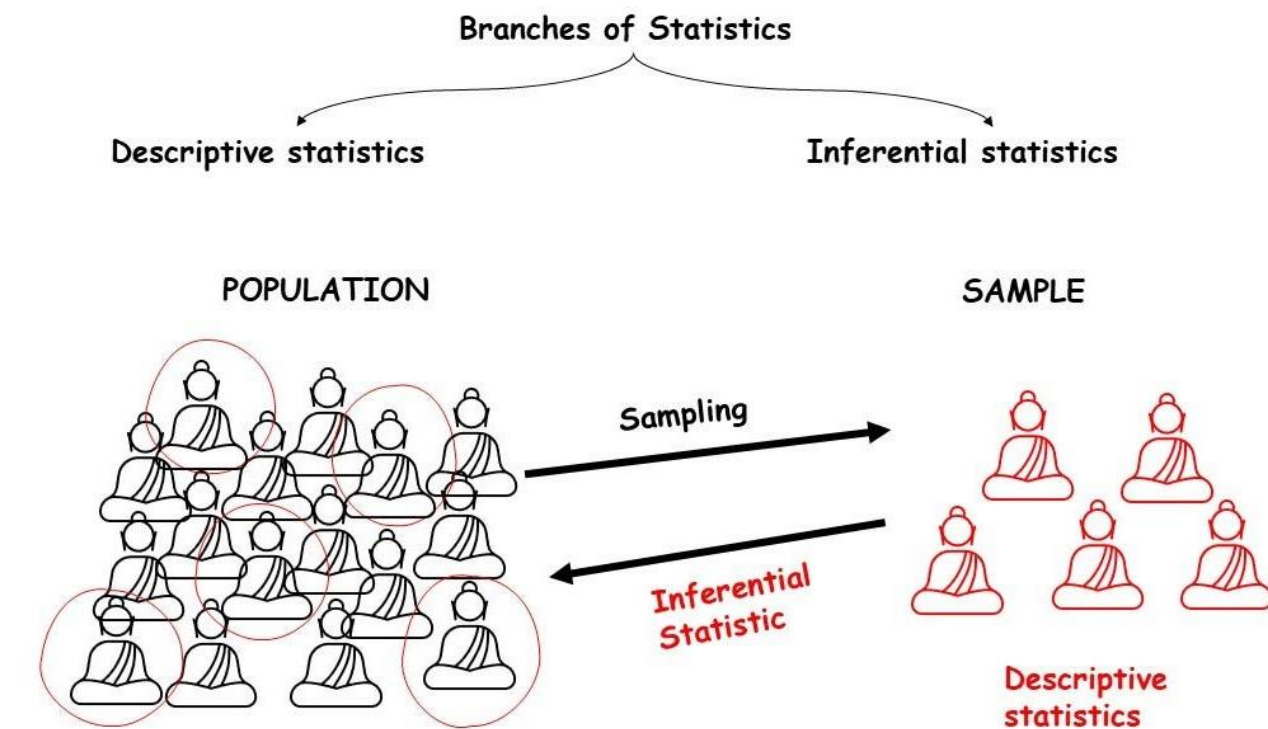
- Summarize observed data.
- Examples: means, percentages, graphs.
- Describe what the sample shows (e.g., 85% said “yes”).

Inferential Statistics

- Use sample data to generalize to the population.
- Produce estimates + uncertainty (e.g., margin of error).
- Example: estimate 83–87% of all adults.

Parameters vs. Statistics

- Parameter = population value (usually unknown).
- Statistic = sample value used to estimate parameter.



We study samples because populations are too large to measure directly



Variables

- All quantitative research relies on variables.
- Research questions = understanding patterns in variables.
- Before analyzing relationships, we need to understand each variable on its own.

What is a variable?

- A variable = a characteristic that varies across subjects.
- A set of observations of the same thing.
- Examples:
 - Income, age, number of siblings.
 - Education level, number of years working in the same job.

Identifying Variable Types

Measurement Scales

- **Nominal**
 - Categories with no order
 - Examples: race/ethnicity, marital status, biological sex, transportation mode
- **Ordinal**
 - Categories with a meaningful order
 - Examples: Likert scales, educational attainment
- **Interval / Ratio (Quantitative)**
 - Numeric values with meaningful differences
 - Interval → no true zero
 - Examples: temperature, dates, IQ score
 - Ratio → true zero
 - Examples: weight, distance, income

Discrete vs. Continuous

- **Discrete variables**
 - Separate, countable values
 - Example: number of children
- **Continuous variables**
 - Infinite possible values
 - Example: age, time, income
- In practice: Variables with many values are often treated as continuous

Why It Matters

- Variables are the foundation of data analysis.
- Correct classification is critical.
- Different types → different methods
- First step: **describe variables before analyzing relationships**

Small Group Exercise

Directions

Work in groups of 4. Using either the NSECE 2019 workforce data or this week's assigned paper, identify examples of variables and classify them.

Then, briefly answer the following:

- Identify at least 4 variables from the dataset and/or paper
- For each variable, determine:
 - Type of variable (measurement scale):
 - Nominal
 - Ordinal
 - Interval/Ratio
 - Form of the variable:
 - Discrete
 - Continuous



Describing Specific Variables

Scale	Type	Display
Nominal	Discrete	Frequency Table
Ordinal	Discrete	Frequency Table
Interval or Ratio	Continuous	Histogram
	Discrete	Frequency Table

Next Class

- Describing variables
- In-class activity with Geller & Curtis (2011)
paper

Questions?

**CLASS 2: DATA,
MEASUREMENT AND
VARIABLES**

Instructor: Aida Pacheco-Applegate
Summer 2026



15-minute break